# Warehouse Indoor Propagation Mini-Project

Geometry-Based RF Modeling for RAIN RFID / Backscatter Environments

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**Abstract**

This mini-project develops a modular MATLAB-based indoor propagation simulator for a warehouse-like environment using 3D scene geometry, ray tracing, and received-power analysis. The goal is to model line-of-sight (LOS), reflections, and diffraction effects in order to understand multipath behavior relevant to RFID reader-tag links in indoor industrial settings. The implementation is structured in an object-oriented way, separating configuration, scene management, simulation, and visualization. The resulting framework supports validation, ablation studies, and future extension toward backscatter-specific modeling.

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## 1 Project Motivation

Indoor RFID deployments such as warehouses, fulfillment centers, and industrial storage spaces are strongly affected by multipath propagation. Reader performance depends not only on transmit power and antenna gain, but also on:

- line-of-sight obstruction,
- reflections from walls, floors, and shelving,
- diffraction around partial blockages,
- tag orientation and placement,
- local fading due to coherent multipath addition.

The purpose of this mini-project is to build a small but physically meaningful simulator that demonstrates how received power changes as the propagation model becomes more realistic.

## 2 Project Objective

The project objective is to create a modular indoor propagation simulation framework that can:

1. import a 3D indoor scene,
2. place a transmitter and multiple receiver probe points,
3. simulate LOS, reflected, and diffracted propagation paths,
4. compute and visualize received signal strength,
5. perform ablation studies on propagation complexity,
6. serve as a foundation for future RFID backscatter modeling.

## 3 System-Level View

The simulation pipeline is:

Scene Geometry → Site Placement → Ray Tracing Model → Path Computation → Received Power / Visualizatio

In the current prototype:

- the transmitter represents an RFID reader antenna,
- the receivers represent tag locations or probe points,
- the environment is modeled using an STL file,
- propagation is simulated using MATLAB ray tracing.

# 4 Modeling Framework

## 4.1 Scene Representation

A 3D STL model is used to represent the indoor environment. For the prototype, the `office.stl` example scene is used because its coordinates are known to be valid and it supports a clean demonstration of obstruction and multipath effects.

## 4.2 Sites

A transmitter site is placed near the ceiling and receiver sites are placed at known valid positions inside the scene. A sweep of receiver points is generated along the straight line between two known-good receiver locations.

## 4.3 Propagation Model

The propagation model is based on MATLAB's ray tracing functionality with:

- Cartesian coordinates,
- configurable surface material,
- controllable number of reflections,
- optional diffraction.

The purpose of this setup is not full-wave electromagnetic modeling, but an engineering-level geometric multipath simulation.

# 5 Mathematical Background

## 5.1 LOS Propagation

For a pure line-of-sight link, the received power follows the Friis transmission law:

$$P_r = P_t G_t G_r \left( \frac{\lambda}{4\pi d} \right)^2$$

where:

- $P_t$ is the transmit power,
- $G_t$ and $G_r$ are the transmitter and receiver antenna gains,
- $\lambda$ is the wavelength,
- $d$ is the Tx-Rx distance.

Thus LOS-only behavior should show smooth distance-dependent decay.

### 5.2 Multipath Propagation

With reflections, the received baseband channel is the coherent sum of multiple paths:

$$h = \sum_{\ell=1}^{L} h_\ell$$

where each path contribution may be written as:

$$h_\ell = \alpha_\ell e^{-j2\pi f_c \tau_\ell}$$

Here:

- $\alpha_\ell$ models spreading loss and reflection effects,
- $\tau_\ell$ is the path delay,
- $f_c$ is the carrier frequency.

The received power is then proportional to:

$$P_r \propto |h|^2$$

This is the key reason multipath can produce both strong peaks and deep fades.

### 5.3 Two-Path Intuition

In the simplest non-LOS extension:

$$h = h_{\text{LOS}} + h_{\text{refl}}$$

If the two paths arrive nearly in phase, they add constructively. If they arrive nearly out of phase, they cancel destructively, producing a deep fade.

### 5.4 Reflection and Diffraction

Adding reflections increases the number of contributing paths. Adding diffraction can allow additional energy to reach points that are partially blocked. In practice:

- reflections create structured fading,
- diffraction can partially fill blocked regions,
- richer propagation models often increase spatial variability.

## 6 Implementation Architecture

The code is written in a modular object-oriented style to match engineering software practices.

### 6.1 Class Structure

The implementation is divided into the following classes:

1. **WarehouseConfig**
   Stores simulation parameters such as scene file, transmitter position, receiver positions, surface material, and ablation settings.

2. **SiteFactory**
   Creates transmitter and receiver site objects, including sweep receivers.

3. **WarehouseScene**
   Owns the scene viewer, site placement, LOS visualization, and map clearing.

4. **PropagationSimulator**
   Owns the ray tracing model and provides methods to configure reflections/diffractions, trace rays, and compute signal strength.

5. **ResultVisualizer**
   Handles plotting of signal strength and ablation results.

6. **main.m**
   Orchestrates the complete experiment.

### 6.2 Why This Structure Matters

This separation provides:

- clean interfaces between environment, simulation, and plotting,
- easy extension to warehouse STL files,
- support for testability and validation,
- a natural path toward more advanced models such as backscatter links.

## 7 Experiment Design

### 7.1 Ablation Study

The main experiment compares three propagation cases:

1. **LOS only**
   No reflections, no diffraction.

2. **One reflection**
   Adds a dominant reflected path.

3. **One reflection + one diffraction**
   Adds both reflected and diffracted energy contributions.

### 7.2 Receiver Sweep

A sequence of receiver points is created between two known-good locations in the STL model. This allows spatial variation in received signal strength to be observed without risking out-of-bounds geometry errors.

### 7.3 Outputs

The simulation produces:

- LOS path visualization,
- ray-traced multipath visualization,
- received signal strength versus sweep position,
- ablation curves comparing propagation complexity.

## 8 Observed Behavior and Interpretation

The key observations from the ablation plot are:

### 8.1 LOS Only

The LOS-only curve is smooth and slowly decreasing. This is expected because only a single path is present and the received power follows the basic free-space trend.

### 8.2 One Reflection

When a reflected path is added, large dips can appear. These occur because the received field is the coherent sum of the LOS and reflected paths:

$$P_r \propto |h_{\mathrm{LOS}} + h_{\mathrm{refl}}|^2$$

At some receiver positions the two paths interfere destructively, causing deep fades.

### 8.3 One Reflection + One Diffraction

When diffraction is also enabled, the received signal contains more terms:

$$P_r \propto |h_{\mathrm{LOS}} + h_{\mathrm{refl}} + h_{\mathrm{diff}}|^2$$

This can both strengthen the average signal and increase spatial fluctuations. In practice, this corresponds to more realistic indoor multipath behavior.

## 9 Simulation Results

This section presents the results obtained from the indoor propagation simulation. The figures illustrate the propagation geometry, ray-tracing behavior, and the effect of multipath complexity on received signal strength.

### 9.1 Indoor Scene and Site Placement

Figure 1 shows the imported STL scene representing the indoor environment. A transmitter site is placed near the center of the room, while two receiver sites are positioned at known valid locations inside the geometry.

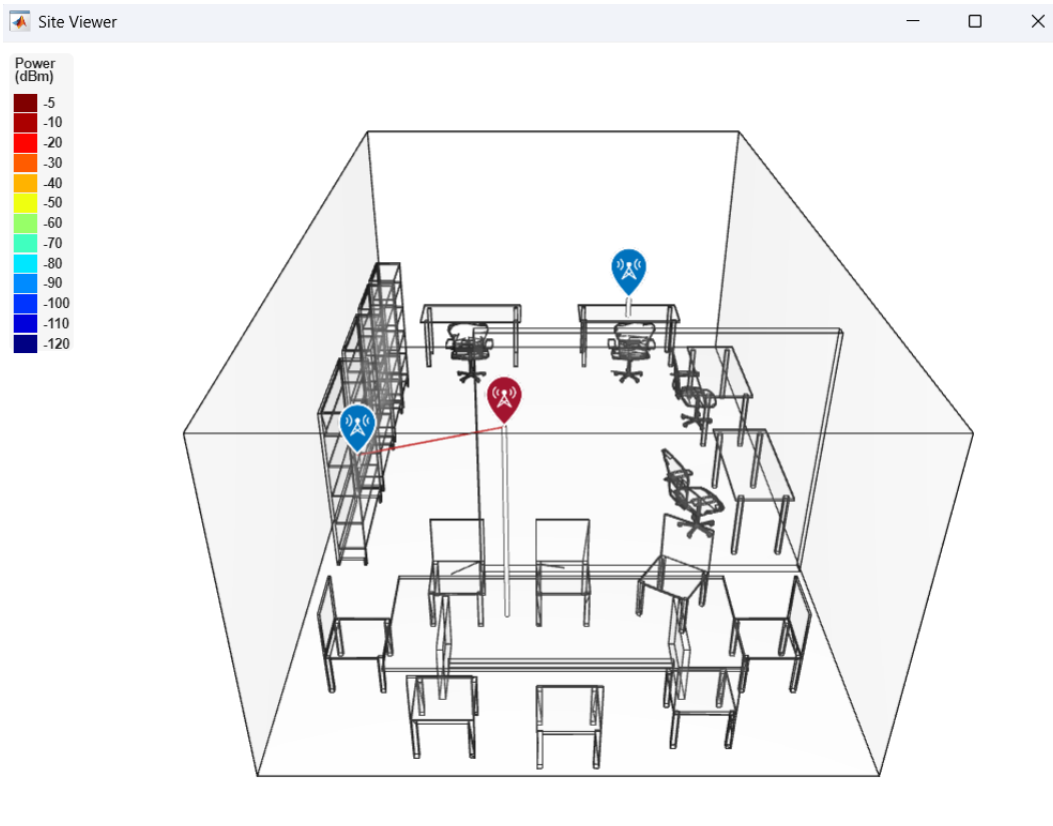


Figure 1: 3D indoor scene imported from an STL file with transmitter and receiver locations. The transmitter (red marker) represents an RFID reader antenna, while the receivers (blue markers) represent tag or probe locations.

This environment serves as the base geometry for all propagation experiments.

### 9.2 Ray-Tracing Propagation Paths

Figure 2 shows the ray-traced propagation paths between the transmitter and receivers when reflections and diffraction are enabled.

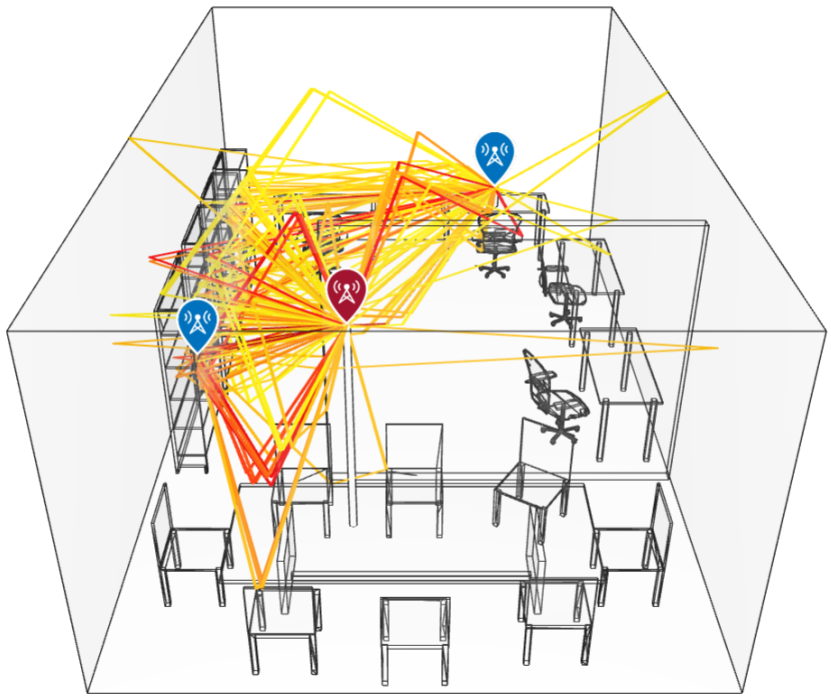

Figure 2: Ray tracing propagation paths with one reflection and one diffraction enabled. Multiple signal paths reach the receiver through reflections and diffraction around obstacles in the environment.

Each colored ray represents a propagation path with a different number of interactions with the environment. The MATLAB site viewer also displays detailed path information including:

- number of reflections
- number of diffractions
- received power
- phase change
- angle of arrival and departure

This allows the simulator to provide insight into how multipath signals combine at the receiver.

### 9.3 Receiver Sweep Experiment

To study how signal strength varies spatially, a sweep of receiver locations was generated between two known receiver points. Figure 3 shows the sweep positions used in the experiment.

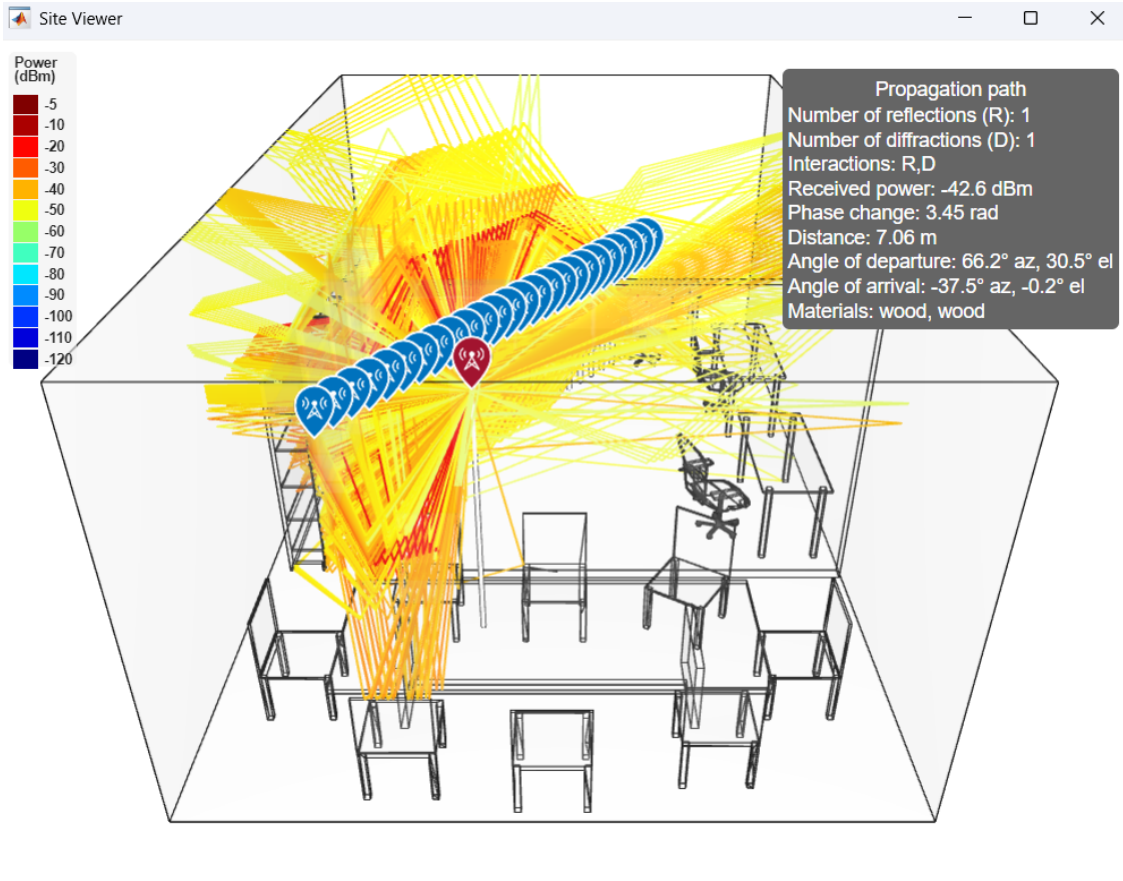


Figure 3: Receiver sweep experiment showing multiple probe locations between two receiver positions. The sweep enables measurement of signal strength variation as a function of spatial position.

This setup allows analysis of spatial fading effects caused by multipath propagation.

### 9.4 Signal Strength Along Receiver Sweep

Figure 4 shows the received signal strength measured along the receiver sweep.

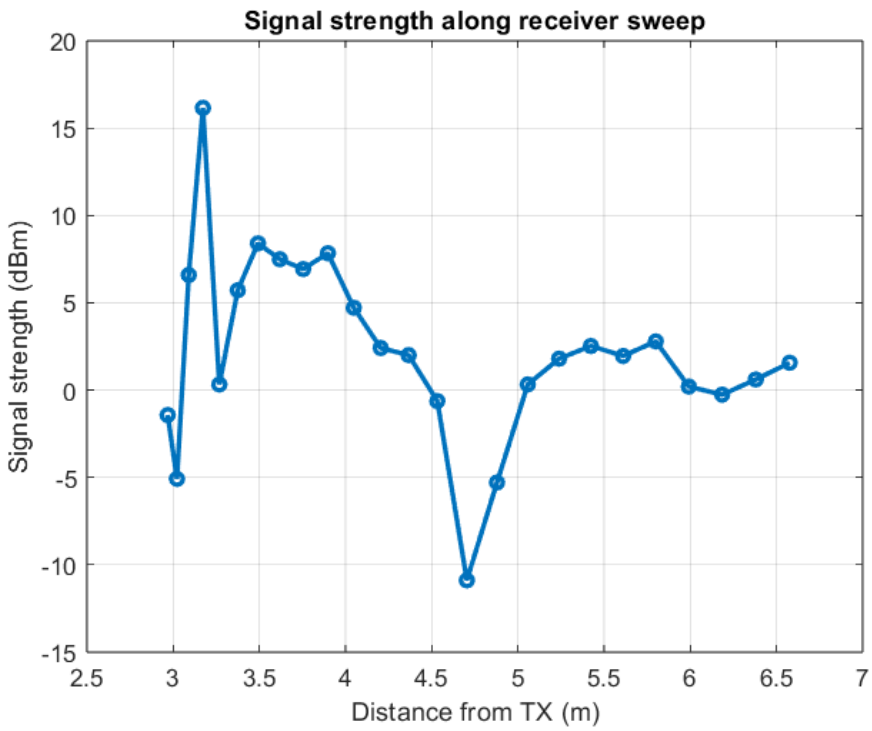


Figure 4: Received signal strength measured along the receiver sweep. Significant fluctuations occur due to constructive and destructive interference between propagation paths.

Several important phenomena can be observed:

- strong peaks caused by constructive interference
- deep fades caused by destructive interference
- spatially varying multipath effects

These behaviors are typical in indoor wireless environments and are especially relevant for RFID deployments in warehouses.

### 9.5 Multipath Ablation Study

To understand the effect of increasing propagation complexity, an ablation study was performed comparing three propagation models:

1. Line-of-sight only
2. One reflection
3. One reflection plus one diffraction

Figure 5 shows the resulting signal strength comparison.

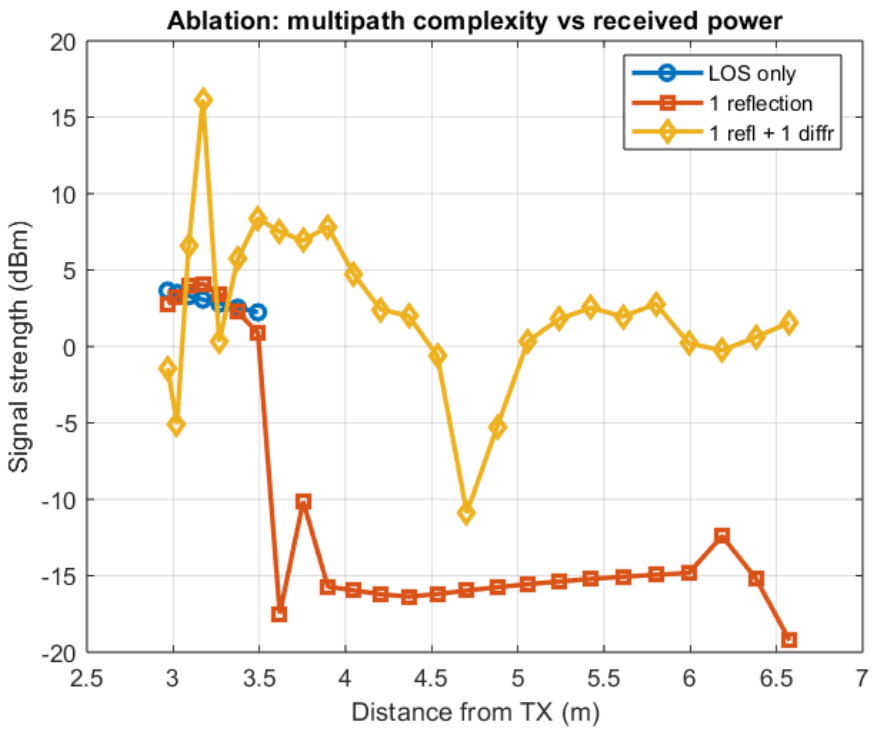


Figure 5: Ablation study showing the effect of multipath complexity on received signal strength. LOS-only propagation shows smooth behavior, while reflections and diffraction introduce strong fluctuations due to interference between multiple paths.

Key observations include:

- LOS-only propagation produces a smooth power trend.
- Adding reflections introduces destructive interference leading to deep fades.
- Adding diffraction introduces additional paths that change the spatial power distribution.

These results highlight the importance of multipath modeling when designing indoor wireless systems.

## 10 Discussion

The simulation results demonstrate that indoor RF propagation is highly sensitive to environmental geometry. Even a small number of reflections can dramatically alter the received signal strength.

The ablation results clearly show that:

- simple LOS models underestimate fading behavior,
- reflections create structured multipath interference,
- diffraction enables signal paths around obstacles.

For RFID systems, these effects influence:

- tag readability
- reader placement
- antenna orientation
- system reliability in cluttered environments

Thus, geometry-based ray tracing provides a useful tool for analyzing indoor wireless system performance.

## 11 Relevance to RFID / Backscatter Systems

Although the current prototype models one-way propagation, it is directly relevant to RFID system design.

A passive RFID system has two important links:

1. **Forward link:** reader → tag
2. **Reverse link:** tag backscatter → reader

The current simulator helps estimate how indoor geometry affects field strength at tag locations. This is important because:

- insufficient forward-link power may prevent tag activation,
- deep fades may reduce read reliability,
- reader placement and antenna orientation affect coverage,
- multipath can strongly influence backscatter detection performance.

A natural future extension is to model the reverse backscatter link using a two-way propagation framework.

## 12 Future Extensions

This framework can be extended in several useful directions:

1. **Warehouse STL replacement**
   Replace `office.stl` with a warehouse model containing aisles and metal racks.
2. **Backscatter modeling**
   Add reader-to-tag and tag-to-reader propagation to estimate effective backscatter power.
3. **Material calibration**
   Use different surface materials and compare their effect on coverage.
4. **Coverage heatmaps**
   Generate 2D or 3D power maps rather than only line sweeps.
5. **Validation suite**
   Create automated tests for LOS sanity, reflection behavior, and reproducibility.
6. **Tag orientation and polarization**
   Include orientation-dependent gain and mismatch effects.

## 13 Conclusion

This mini-project demonstrates a practical and modular approach to indoor RF propagation simulation in MATLAB. Starting from a 3D scene model and a small set of valid transmitter and receiver locations, it shows how LOS, reflections, and diffraction affect received power. The ablation study highlights the central role of coherent multipath in creating spatial fading, and the object-oriented software structure makes the code easier to extend, test, and discuss in an engineering interview.

Overall, this project serves as a compact but relevant prototype for indoor RFID channel modeling and provides a strong foundation for future work on warehouse-scale propagation and backscatter-aware system simulation.